iDARVP study design and evaluation workflow

Homology-aware benchmarking and analysis framework with a two-stage case-study implementation

1. Data curation and split design

Integrated evidence

- ARVP-positive evidence from curated peptide resources
- Negative-A: APD6-pool background peptides
- Negative-B: non-Retroviridae antiviral stress set

Homology-aware partitioning

- Stage 1: CD-HIT 80% cluster-level assignment
- Negative-B-containing clusters fixed in test_hard
- Stage 2: stricter hom40 cluster-aware split

Traceability and uncertainty

- Preserved split files and provenance records
- Validation-only threshold selection
- Cluster bootstrap + repeated-assignment sensitivity

2. Stage 1 ARVP identification

Training and validation

- Balanced ARVP positives + Negative-A
- Separate train and validation sets
- Controlled kmer13 and ESM2-t6 baselines

Easy vs hard evaluation

- test_easy: matched-background balanced holdout
- test_hard: positives + annotation-based Negative-B
- AUPRC, AUROC, MCC, calibration diagnostics

Ranking-oriented analysis

- Top-K at K = 50, 100, 200
- Precision@K, Recall@K, Enrichment@K
- Directional interpretation when intervals overlap

3. Stage 2 mechanism prediction

Stage 2 input and labels

- ARVP sequences under hom40 control
- Six biological activity categories
- MAP*: analytical multi-activity grouping

Model comparison

- kmer13 baseline and ESM2-t6 embedding baseline
- kmer13 + ESM2 fusion variants
- One-vs-rest logistic regression

Evaluation

- Per-label AUPRC and threshold-dependent metrics
- Validation-only per-label threshold calibration
- Rare-label results interpreted conservatively

Supplementary exploratory branch

AI4AVP / AVPpred historical pilot retained as descriptive context only; paired per-sequence outputs were not preserved, so no paired superiority claim is made.

Reproducibility release: curated splits • scripts • thresholds • robustness outputs

Take-home: apparent performance depends on benchmark design, homology control, and negative-set definition.

** MAP is an analysis-specific multi-activity grouping, not a biological mechanism.*